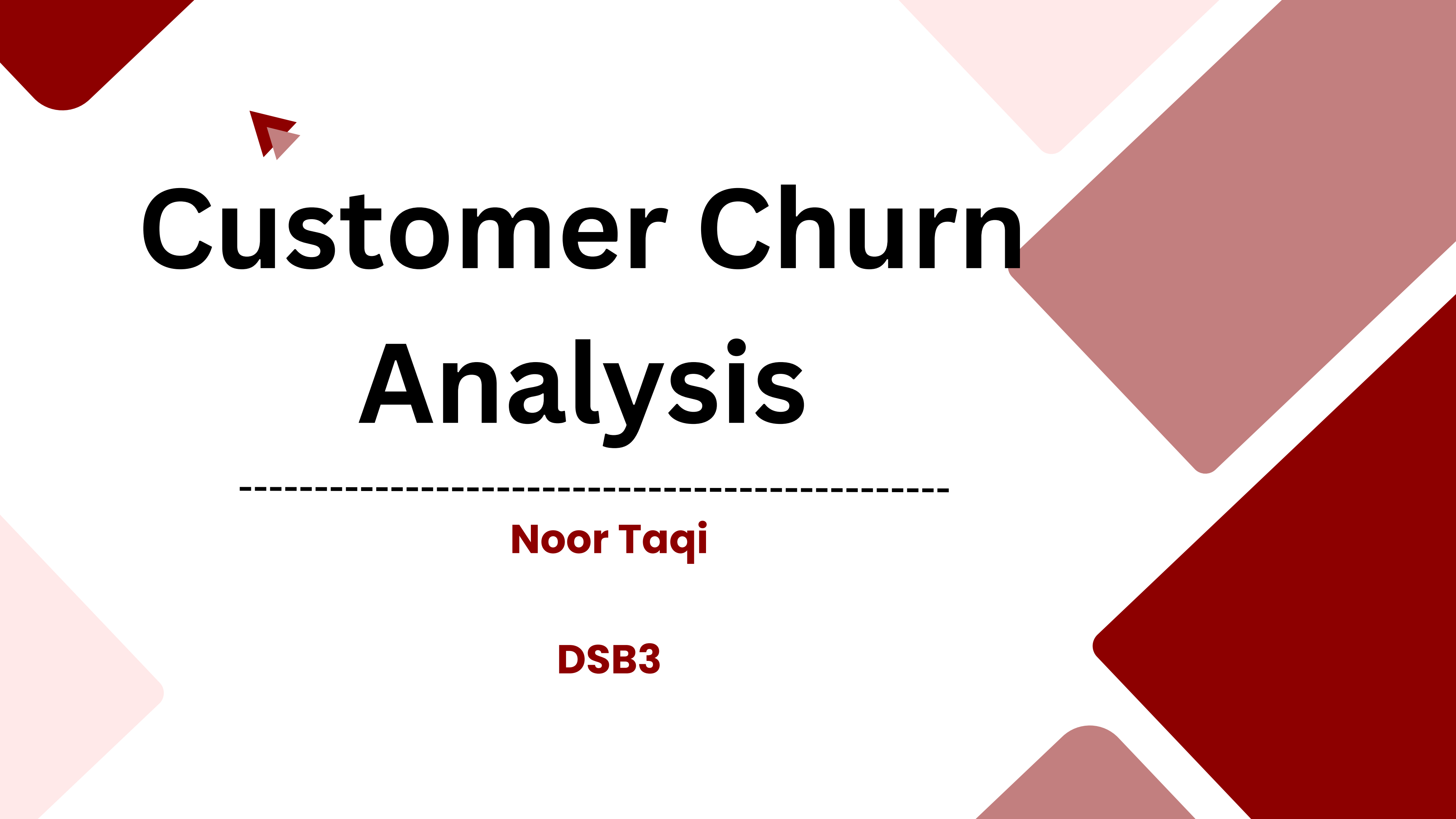




Customer Churn Analysis

Noor Taqi

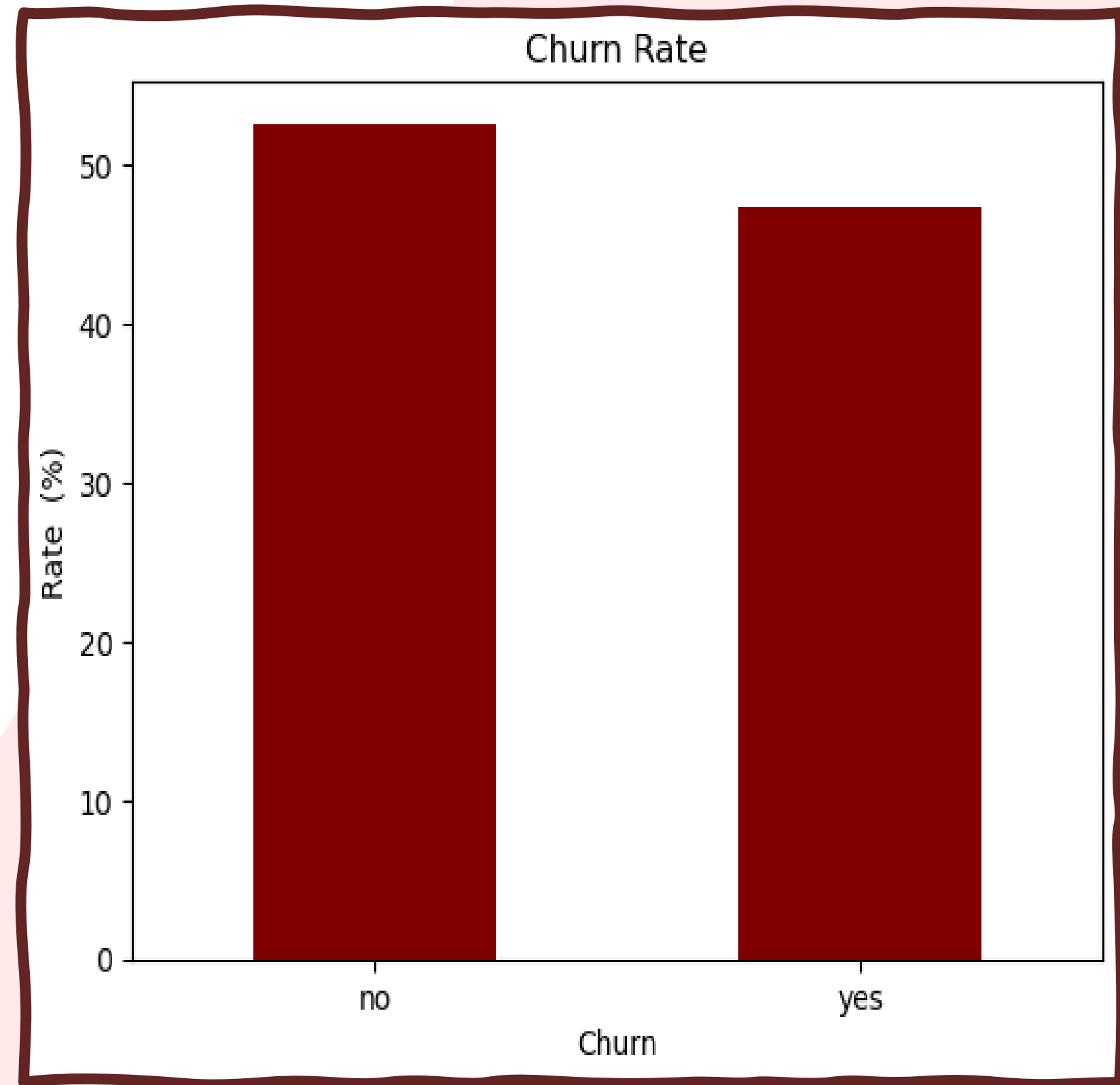
DSB3

Agenda

- Problem Statement
- Aim
- Objectives
- Data Overview
- Analysis
- Recommendation

Problem statement

The company is facing a severe customer churn crisis, with **47.4%** of 64,374 subscribers cancelling their service, resulting in an estimated **\$18,991,722** in lost revenue and substantially higher customer acquisition costs.



Aim

Identify the key drivers of customer churn and provide data-driven strategies to improve customer retention and protect recurring revenue.

Objectives



1

What is the churn rate across age categories and gender, and which age-gender segment has the highest churn rate?

2

How does the churn rate vary by contract length (Monthly, Quarterly, Annual) and subscription type (Basic, Standard, Premium), and which factor differentiates churn more?

3

Which tenure months (1-60) have the highest and lowest churn rates, and does low tenure imply a high churn rate?

4

What percentage of customers had a payment delay exceeding 14 days across all customers, and how does the churn rate differ between high-delay and low-delay customers?

5

How does the churn rate vary with the number of support calls (0-10), and how does it differ between customers with more than 6 calls and those with 6 or fewer?

Data Overview

Total Number of Customers: 64,374 subscribers.

Size: 64,374 customers (rows), 12 columns

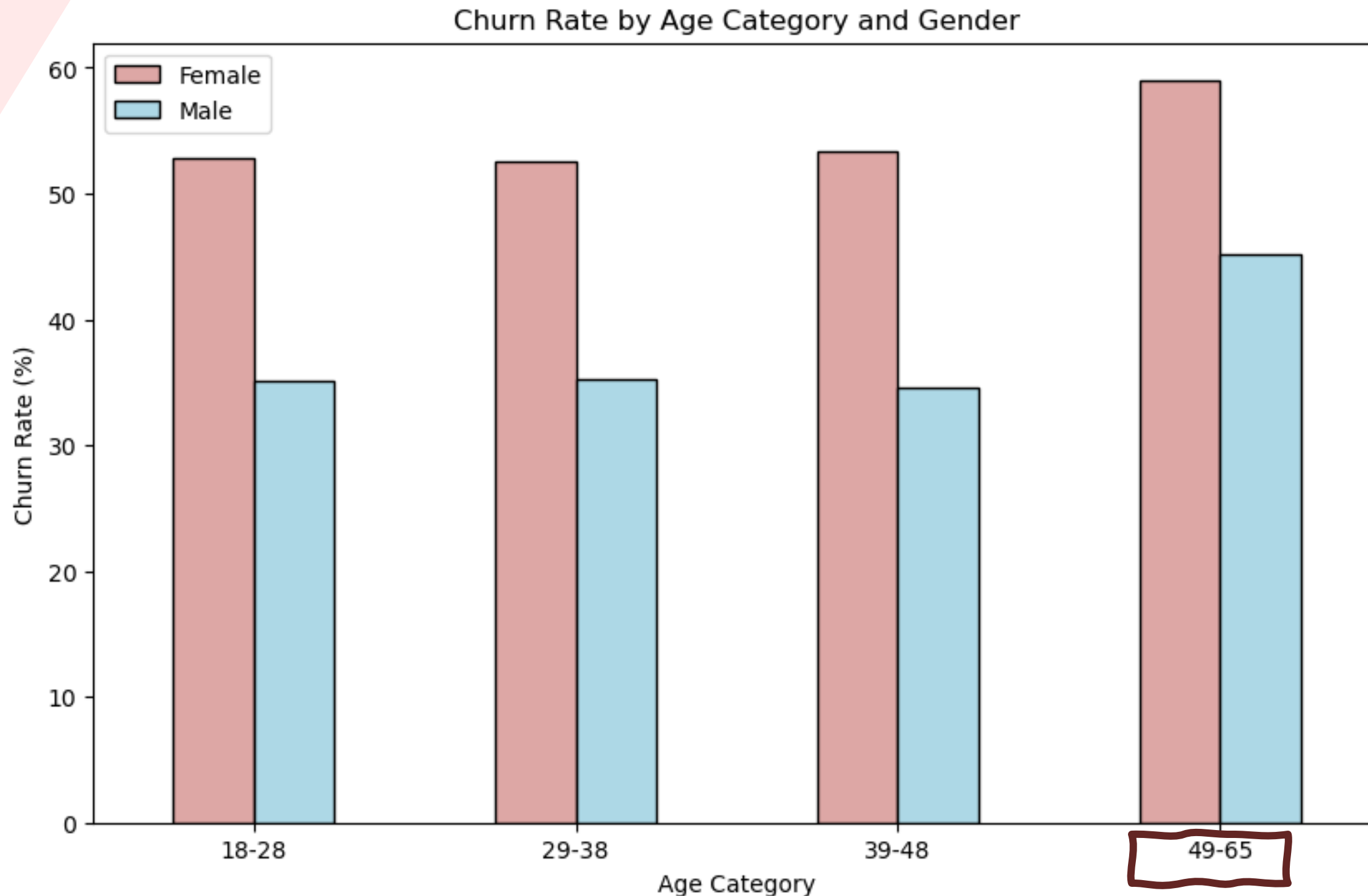
Target variable: Churn (yes/no), 47.4% of customers churned

Data quality: no missing values, no duplicates

Source: Kaggle, Customer Churn Dataset

Analysis

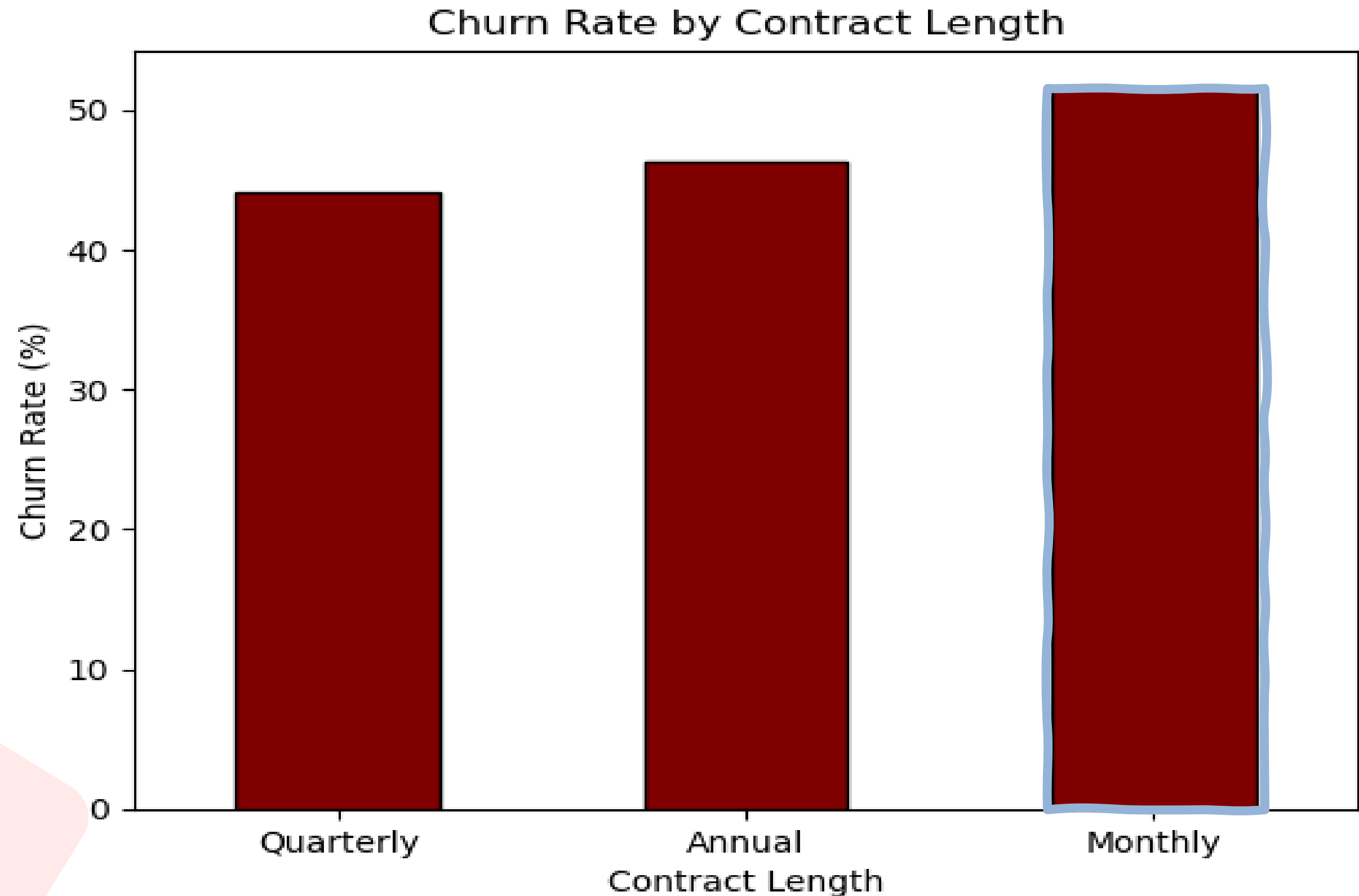
Churn by Age & Gender



Female subscribers exhibit a consistently higher churn rate than **male** subscribers across all age groups. While churn peaks in the **49–65 age category** for both genders.

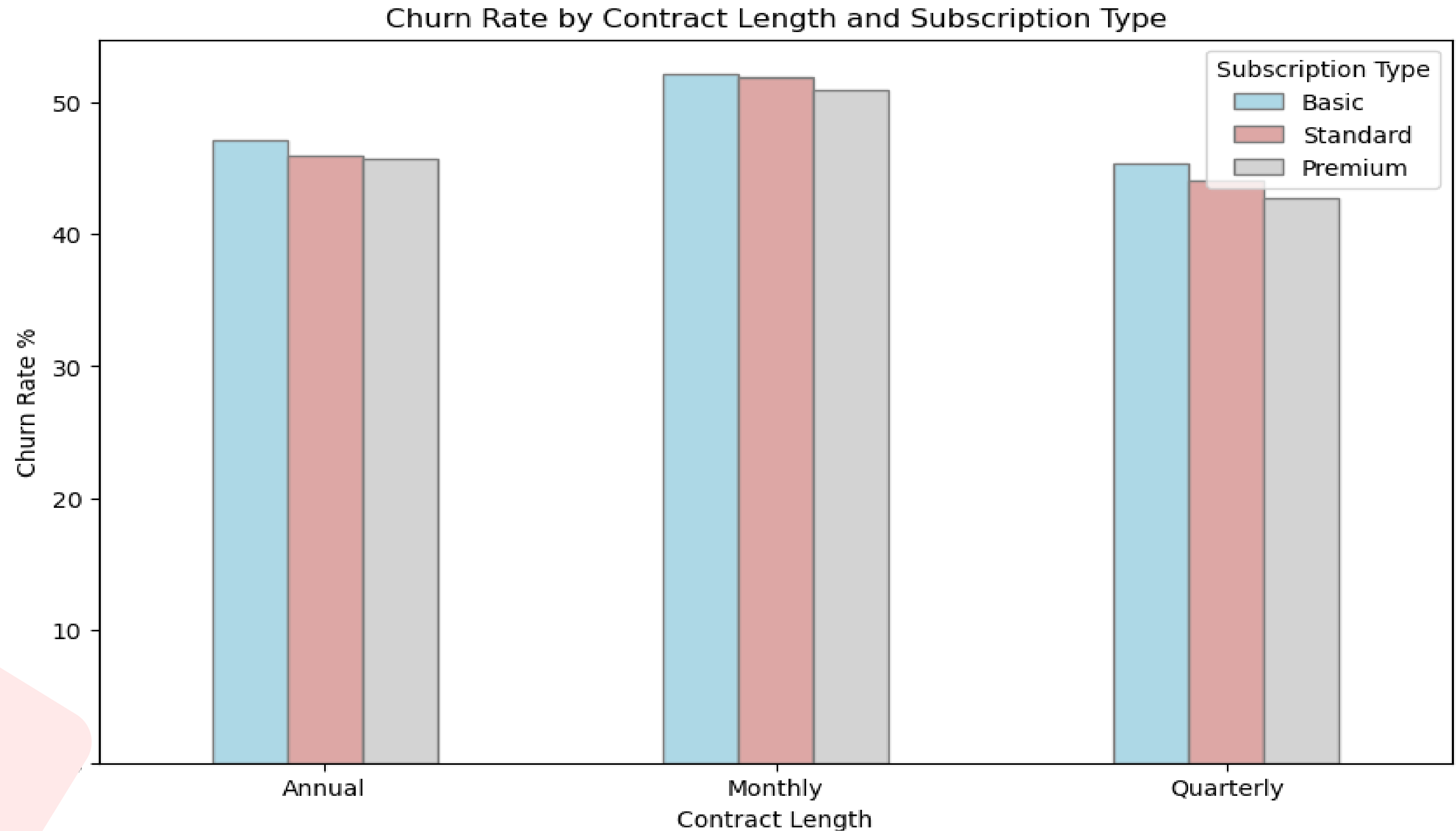
Contract Length Impact

Monthly subscribers record the highest churn rate at over **50%**, This confirms that **short-term** contracts are a main **driver of customer loss**.



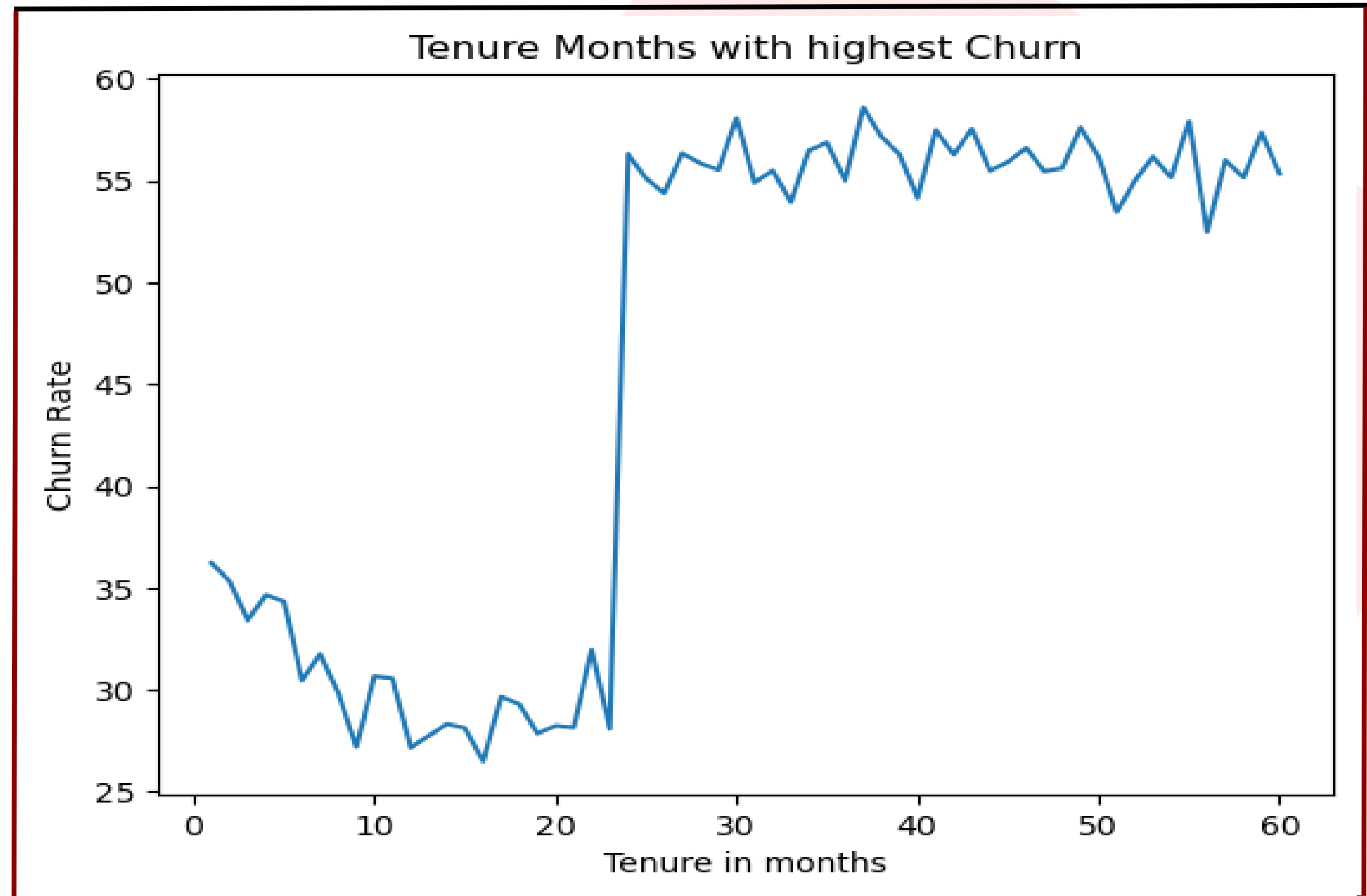
Contract Length Impact

Subscription type has negligible impact on churn, with the **Basic plan** recording only slightly higher rates .



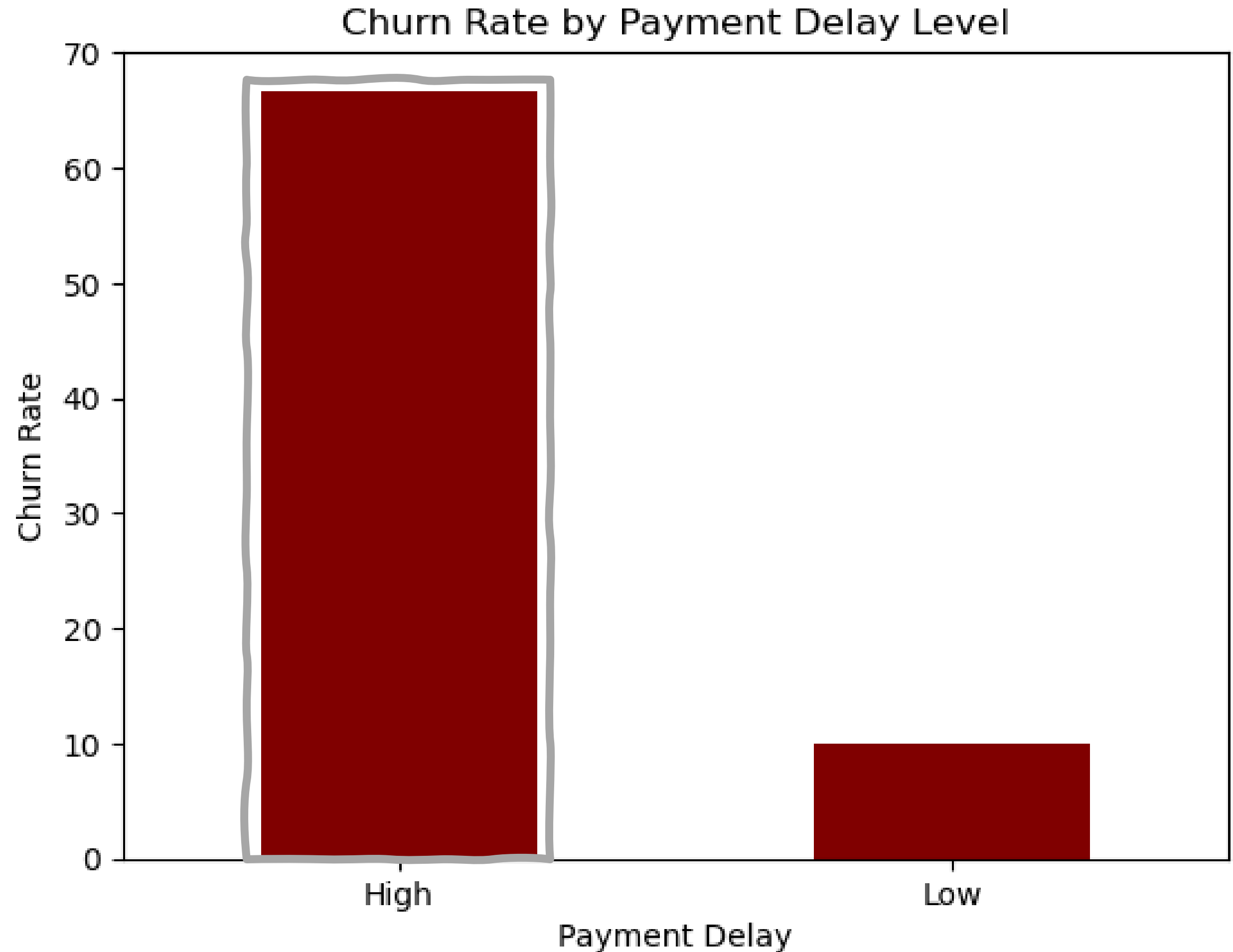
Customer's Tenure

- Customer tenure splits into two distinct groups, jumping from lowest churn under 22 months to highest after.
- Higher tenure directly correlates with a higher churn rate.



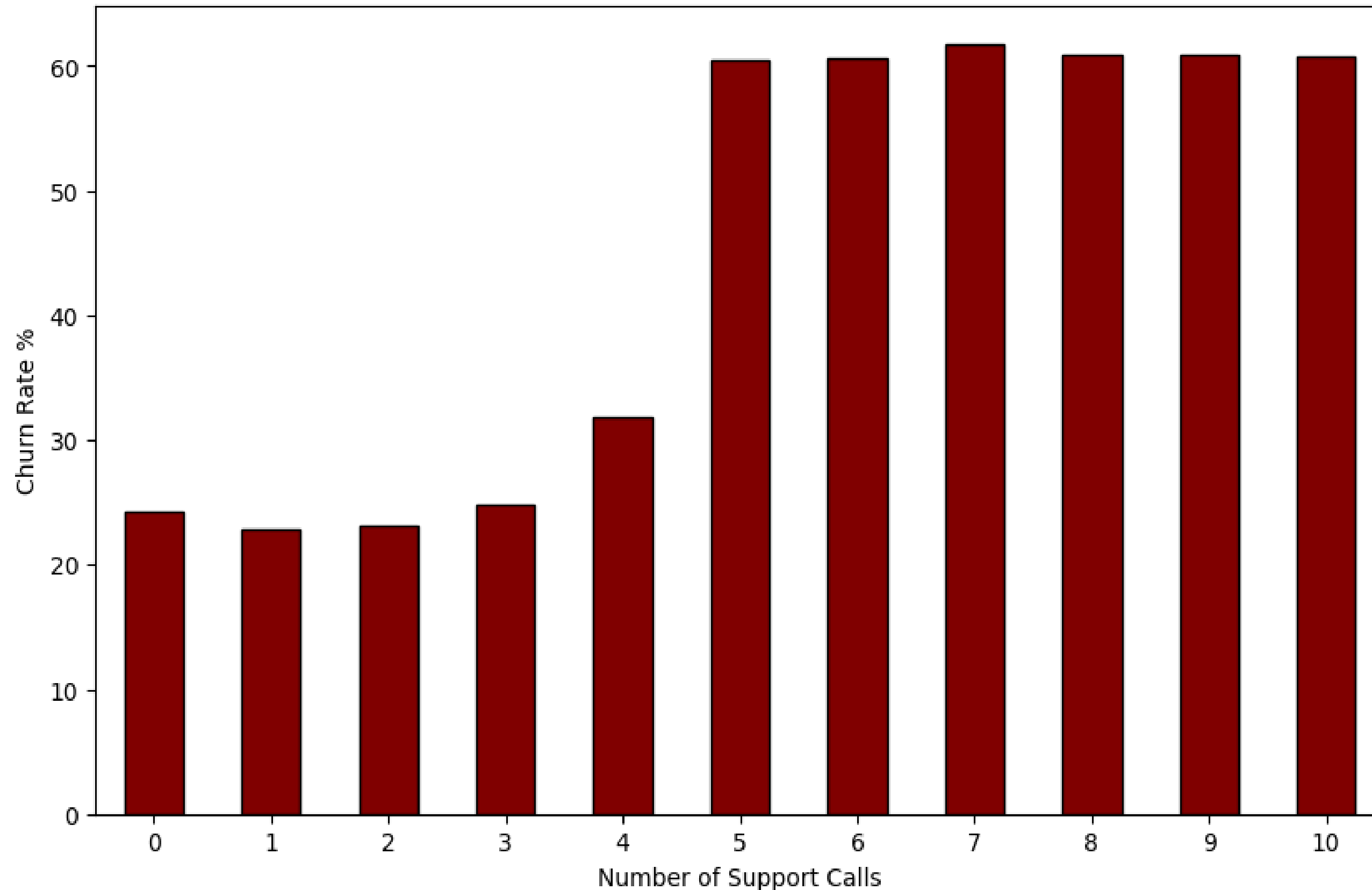
Churn Risk by Payment Delay

Customers with **high payment delays** are nearly **five times** more likely to churn than those with **low delays**.



Support calls **Threshold**

Churn Rate by Number of Support Calls



- Churn rate spikes sharply to **over 60%** after **4** support calls.
- Frequent calls may **indicate service issues**, leading to customer churn.

Recommendation

- Monitor customers who make frequent support calls and follow up on existing issues.
- Encourage customers to switch to longer-term contracts, such as annual ones.
- Offer flexible payment options and reminders to customers who are more likely to be late with payments.
- Offer tiered upgrades based on tenure, giving long-standing customers added value that discourages them from cancelling their subscription.

The background features two large, light pink geometric shapes. One is a triangle on the left side, and the other is a larger, more complex shape on the right side, both pointing towards the center.

THANK YOU